

where r_{it} is the return of CalPERS, r_{st} is the S&P 500 equity market return, and r_{bt} is a bond portfolio return—in this case, the Ibbotson Associates long-term corporate bond total return index. To obtain a benchmark portfolio, we require the restriction that

$$\beta_s + \beta_b = 1.$$

That is, the portfolio weights must sum to one. Then, \$1 placed into CalPERS on the left-hand side of equation (10.12) can be replicated by a portfolio of stocks and bonds (with portfolio weights, which also must sum to one) on the right-hand side, plus any alpha generated by the CalPERS’ funds manager.

Estimating equation (10.12) on CalPERS’ annual returns from 1990 to 2011, we get the following:

	Coefficient	T-stat
Alpha	-1.11%	-1.16
Bond Loading	0.32	13.97
Stock Loading	0.68	13.97
Adj R^2	0.90	

The high adjusted R^2 of 90% is amazing: CalPERS’ returns are extremely well explained by this mimicking portfolio of 32% bonds and 68% stocks!

The point estimate of CalPERS alpha is negative, at -1.11% per year. Should we immediately fire the CalPERS funds manager and put everything into low-cost index funds? Formally, we can only make the statement that “we fail to reject the hypothesis that CalPERS adds value relative to the 32% bonds/68% stocks benchmark portfolio at the 95% level” because the t -statistic is less than two in absolute value.

CalPERS, however, is an expensive fund. In 2011 its internal estimate of its expense ratio was upward of 0.50%, while expense ratios inferred from its annual reports exceed 0.80%. (What a travesty that it does not explicitly report its expense ratio in its annual report!) These expense ratios are much higher than those of industry peers. The median expense ratio of the largest pension plans studied by Bauer, Cremers, and Frehen (2009) was 0.29%; at the largest 30% of defined benefit plans, the expense ratios are just 0.15%. Thus, CalPERS is three to four times more expensive than median fund in Bauer, Cremers, and Frehen’s sample, and nearly nine times more expensive than the largest 30% of pension plans! Expense ratios for managing typical index stock or bond funds at CalPERS’ scale are way below 0.10%. (The Norwegian sovereign wealth fund had an expense ratio of 0.06% in 2012.) So yes, given that CalPERS could run a benchmark stock-bond portfolio for close to zero, perhaps it should consider firing its funds manager and going completely index.